

ACID PROPERTIES AND ISOLATION

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Create a database & use:

```
1 CREATE DATABASE IF NOT EXISTS bookflow2_db;
2 USE bookflow_db;
```

Output

#	Time	Action
1	09:08:46	SHOW DATABASES
2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db
3	09:09:40	USE bookflow_db

Create a table:

```
3 CREATE TABLE Customer (
4     Customer_ID INT PRIMARY KEY,
5     Customer_Name VARCHAR(100) NOT NULL,
6     Phone VARCHAR(15),
7     City VARCHAR(50)
8 );
9
```

Output

#	Time	Action
1	18:35:06	CREATE DATABASE IF NOT EXISTS bookflow12_db
2	18:35:26	USE bookflow12_db
3	18:35:50	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY,

```
9 CREATE TABLE Account (
10     Account_No INT PRIMARY KEY,
11     Customer_ID INT,
12     Account_Type VARCHAR(20),
13     Balance DECIMAL(12,2),
14     Branch VARCHAR(50),
15
16     FOREIGN KEY (Customer_ID)
17     REFERENCES Customer(Customer_ID)
18 );
19
```

```
19 CREATE TABLE Bank_Transaction (
20     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
21     Account_No INT,
22     Transaction_Type VARCHAR(20),
23     Amount DECIMAL(12,2),
24     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
25
26     FOREIGN KEY (Account_No)
27     REFERENCES Account(Account_No)
28 );
29
```

Output

#	Time	Action
3	18:35:50	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer
4	18:36:33	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer
5	18:36:34	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer
6	18:36:39	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer,
7	18:37:29	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY,

Insert data into customer:

```
29 * INSERT INTO Customer
30 (Customer_ID, Customer_Name, Phone, City)
31 VALUES
32 (101, 'Lakshmi', '9876543210', 'Hyderabad'),
33 (102, 'Lakshmi', '9876543211', 'Vijayawada'),
34 (103, 'Lakshmi', '9876543212', 'Bangalore'),
35 (104, 'Tejas', '9876543213', 'Chennai'),
36 (105, 'Tejuu', '9876543214', 'Hyderabad');
```

#	Time	Action
4	18:36:33	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(50), Phone VARCHAR(15), City VARCHAR(50))
5	18:36:34	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(50), Phone VARCHAR(15), City VARCHAR(50))
6	18:36:35	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(50), Balance DECIMAL(10,2), Branch VARCHAR(50))
7	18:37:29	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction_Type VARCHAR(50), Amount DECIMAL(10,2))
8	18:38:55	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Lakshmi', '9876543210', 'Hyderabad')

Insert into account:

```
* INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 80000, 'Hyderabad');
```

#	Time	Action
5	18:36:34	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(50), Phone VARCHAR(15), City VARCHAR(50))
6	18:36:35	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(50), Balance DECIMAL(10,2), Branch VARCHAR(50))
7	18:37:29	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction_Type VARCHAR(50), Amount DECIMAL(10,2))
8	18:38:55	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Lakshmi', '9876543210', 'Hyderabad')
9	18:39:57	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyderabad')

Insert into transaction:

```
45 * INSERT INTO Bank_Transaction
46 (Account_No, Transaction_Type, Amount)
47 VALUES
48 (10001, 'DEPOSIT', 10000),
49 (10002, 'DEPOSIT', 15000),
50 (10003, 'WITHDRAW', 20000),
51 (10004, 'DEPOSIT', 5000),
52 (10005, 'WITHDRAW', 10000);
```

#	Time	Action
6	18:36:39	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(50), Balance DECIMAL(10,2), Branch VARCHAR(50))
7	18:37:29	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction_Type VARCHAR(50), Amount DECIMAL(10,2))
8	18:38:55	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Lakshmi', '9876543210', 'Hyderabad')
9	18:39:57	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyderabad')
10	18:40:32	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)

Select:

#	Time	Action
53	18:40:32	SELECT * FROM Customer;
54	18:40:32	SELECT * FROM Account;
55	18:40:32	SELECT * FROM Bank_Transaction;

Customer_ID	Customer_Name	Phone	City
101	Lakshmi	9876543210	Hyderabad
102	Lakshmi	9876543211	Vijayawada
103	Lakshmi	9876543212	Bangalore
104	Tejas	9876543213	Chennai
105	Tejuu	9876543214	Hyderabad

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	80000.00	Hyderabad

```
55 * SELECT * FROM Bank_Transaction;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2020-09-18 10:40:32
2	10002	DEPOSIT	10000.00	2020-09-18 10:40:32
3	10003	WITHDRAWAL	20000.00	2020-09-18 10:40:32
4	10004	DEPOSIT	10000.00	2020-09-18 10:40:32
5	10005	WITHDRAWAL	10000.00	2020-09-18 10:40:32

```
56 * START TRANSACTION;
```

```
57
```

#	Time	Action
13	19:06:41	SELECT * FROM Bank_Transaction LIM
14	19:07:19	START TRANSACTION

Deposit money:

```
57 * UPDATE Account
```

```
58 SET Balance = Balance + 5000
```

```
59 WHERE Account_No = 10001;
```

#	Time	Action
14	19:07:19	START TRANSACTION
15	19:09:43	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001

Check balance:

```
60 * SELECT *
```

```
61 FROM Account
```

```
62 WHERE Account_No = 10001;
```

```
63
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

Withdraw money:

```
64 * UPDATE Account
```

```
65 SET Balance = Balance - 10000
```

```
66 WHERE Account_No = 10001;
```

#	Time	Action
17	15:11:29	COMMIT
18	15:11:42	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001

Check balance:

```
67 * SELECT *
```

```
68 FROM Account
```

```
69 WHERE Account_No = 10001;
```

```
70
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad

First Update:

```
71 * UPDATE Account
72 SET Balance = Balance + 5000
73 WHERE Account_No = 10001;
```

Output

Action Output

#	Time	Action
20	19:13:06	ROLLBACK
21	19:13:40	UPDATE Account SET Balance = Balance +

Save deposit:

```
74 * SAVEPOINT Deposit1;
75
```

Output

Action Output

#	Time	Action
21	19:13:40	UPDATE Account SET Balance = Ba
22	19:14:23	SAVEPOINT Deposit1

Second update:

```
75 * UPDATE Account
76 SET Balance = Balance - 3000
77 WHERE Account_No = 10002;
```

Output

Action Output

#	Time	Action
22	19:14:23	SAVEPOINT Deposit1
23	19:14:58	UPDATE Account SET Balance = Balance - 3000 V

Third update:

```
79 * UPDATE Account
80 SET Balance = Balance + 10000
81 WHERE Account_No = 10003;
```

Output

Action Output

#	Time	Action
24	19:15:36	SAVEPOINT Withdrawal1
25	19:15:38	UPDATE Account SET Balance = Balance + 10000

Check accounts:

```
85 FROM Account;
```

result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	72000.00	Vijayawada
10003	103	Current	130000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad